

Partial Differential Equations

Partial Differential Equation (PDE) an equation involving an unknown multivariate function and its partial derivatives.

- PDE can model a wide variety of phenomena.

General theory of solving all PDE:

unknown; extremely unlikely to exist.

- Weak solution and regularity:

Certain specific PDE (e.g., Laplace's equation):

can be solved classically;

Many others, if not most others: cannot.

- Existence & uniqueness:
(arbitrary functions from integration, e.g., $\frac{\partial \sigma}{\partial y} = 2y$.)
B.C./I.C.: sufficient, but not too much? generally hard!
- Order of PDE: highest order of derivatives.
- Linear PDE: no powers of u and its derivatives.
 $Lu = f$ where L is a linear differential operator.
Principle of superposition.

主要内容

例子或类型：解的表示或性质

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解法

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分析方法

.....

Four Linear Fundamental PDEs to study

- Transport equation: $u_t + \mathbf{b} \cdot D_x u = 0$.
- Laplace's equation: $\Delta u = 0$.
- Heat equation: $u_t - \Delta u = 0$.
- Wave equation: $u_{tt} - \Delta u = 0$.

Various explicit formulas exist!

Prerequisite: integration by parts, Green's formula, convolution, etc.

Transport Equation

($\rightarrow$ ODE: method of characteristics.)

- Initial value problem

$$\begin{cases} u_t + \mathbf{b} \cdot D_x u = 0, & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

- $g \in C^1$: classical solution.
- $g \notin C^1$: a need for weak solution!

- Nonhomogeneous problem

$$\begin{cases} u_t + \mathbf{b} \cdot D_x u = f, & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

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Laplace's/Poisson's Equation

(among the most important of all PDEs)

Laplace's equation $\Delta u = 0$. ($u \in C^2$: harmonic.)

Poisson's equation $-\Delta u = f$.

Here $\Delta u = \sum_{i=1}^n u_{x_i x_i}$

- u : chemical concentration, temperature, potential, etc.
in equilibrium with/without source.
- Also arise in other areas, e.g., Brownian motion, etc.

1. Fundamental Solution $\Phi(x)$ (radial solution in $U = \mathbb{R}^n$)

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & n = 2 \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}}, & n \geq 3 \end{cases}$$

$\alpha(n)$: volume of the unit ball in $\mathbb{R}^n$.

($|B(r)| = \alpha(n)r^n$, $|\partial B(r)| = n\alpha(n)r^{n-1}$.)

$-\Delta\Phi = \delta_0$ in $\mathbb{R}^n$.

$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy$ solves Poisson's equation
 $-\Delta u = f \in C_c^2(\mathbb{R}^n)$ in $\mathbb{R}^n$

2. Mean-value formula $U \subset \mathbb{R}^n$ open,

$$u \in C^2(U) \text{ harmonic} \iff u(x) = \frac{1}{|\partial B(x,r)|} \int_{\partial B(x,r)} u dS$$

for each $B(x,r) \subset U$.

Consequently, $u(x) = \frac{1}{|B(x,r)|} \int_{B(x,r)} u dy.$

3. Maximum Principle $U \subset \mathbb{R}^n$ open and bounded

$u \in C^2(U) \cap C(\bar{U})$, harmonic in U ,

- $\max_{\bar{U}} u = \max_{\partial U} u$;
- U connected; $u(x_0) = \max_{\bar{U}} u, x_0 \in U \Rightarrow u$ constant.
(Strong Max. Principle)

A uniqueness result follows:

at most one $u \in C^2(U) \cap C(\bar{U})$ satisfies

$$-\Delta u = f \in C(U) \text{ in } U, u = g \in C(\partial U) \text{ on } \partial U.$$

4. Regularity and Local Estimates u harmonic in U

- $u \in C^\infty(U)$: η a mollifier, $u_\epsilon = \eta_\epsilon * u \in C^\infty(U_\epsilon)$;
mean value formula $\Rightarrow u = u^\epsilon$ in U_ϵ .
- $|D^\alpha u(x_0)| \leq \frac{(2^{n+1}nk)^k}{\alpha(n)r^{n+k}} \|u\|_{L^1(B(x_0,r))}$, $\forall B(x_0, r) \subset U$,
where $|\alpha| = k$: $D^\alpha u$ harmonic; mean value formula;
mathematical induction.
- u is analytic in U : by the above local estimates.

5. Liouville's Theorem

u harmonic in $\mathbb{R}^n$; $|u| \leq M$ (constant)

$\implies u$ constant.

$$(|Du(x_0)| \leq \frac{C_1}{r^{n+1}} \|u\|_{L^1(B(x_0, r))} \leq \frac{C_1 \alpha(n)}{r} M \rightarrow 0)$$

$f \in C_c^2(\mathbb{R}^n)$, $n \geq 3$.

Any bounded solution of $-\Delta u = f$ in $\mathbb{R}^n$ has the form

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy + C$$

for some constant C .

6. Harnack's Inequality

$V \subset \bar{V} \subset U$ and $\bar{V}$ compact

$\implies \exists C$ (depending on V) s.t. $\sup_V u \leq C \inf_V u$
for all u nonnegative, harmonic in U .

(By the mean value formula.)

Green's Function

- Green's identity: ν outer unit normal on ∂V ,

$$\int_U [u \Delta v + \nabla u \cdot \nabla v] dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} dS,$$

$$\int_U [u \Delta v - v \Delta u] dx = \int_{\partial U} (u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu}) dS.$$

- Green's function to Dirichlet B.C. $\begin{cases} -\Delta u = f \text{ in } U \\ u = g \text{ on } \partial U \end{cases} :$

$$G(x, y): \text{ for fixed } x \in U, \begin{cases} -\Delta_y G(x, y) = \delta_x(y) \text{ in } U \\ G(x, y) = 0 \text{ on } \partial U \end{cases}$$

- $u(x) = \int_U G(x, y) f(y) dy - \int_{\partial U} g(y) \frac{\partial G(x, y)}{\partial \nu} dy.$
(the first part: $g = 0$; the second part: $f = 0$.)
- $G(x, y) = \Phi(y - x) - \phi^x(y), \quad x, y \in U, x \neq y,$
where $\Delta_y \phi^x(y) = 0$ in U ; $\phi^x(y) = \Phi(y - x)$ on ∂U .
- $G(x, y) = G(y, x).$

- Green's function to Neumann B.C. $\begin{cases} -\Delta u = f \text{ in } U \\ \frac{\partial u}{\partial \nu} = g \text{ on } \partial U \end{cases} :$

$$G(x, y): \text{ for fixed } x \in U, \begin{cases} -\Delta_y G(x, y) = \delta_x(y) \text{ in } U \\ \frac{\partial G(x, y)}{\partial \nu} = 0 \text{ on } \partial U \end{cases}$$

- $u(x) = \int_U G(x, y)f(y)dy + \int_{\partial U} G(x, y)g(y)dy.$

(the first part: $g = 0$; the second part: $f = 0$.)

- $G(x, y) = \Phi(y - x) + \text{some harmonic corrector} \dots$

- $G(x, y) = G(y, x).$

- Find $G(x, y)$? Difficult; method of image (special U).

- Half Space $\mathbb{R}_+^n = \{x = (x_1, \dots, x_{n-1}, x_n) : x_n > 0\}$,
 $\tilde{x} = (x_1, \dots, x_{n-1}, -x_n)$.

Dirichlet B.C.: $G(x, y) = \Phi(y - x) - \Phi(y - \tilde{x})$,

Neumann B.C.: $G(x, y) = \Phi(y - x) + \Phi(y - \tilde{x})$,
 $x, y \in \mathbb{R}_+^n$.

- A Ball $B_R = \{x \in \mathbb{R}^n : |x| < R\}$, $\tilde{x} = \frac{R^2}{|x|^2}x$,

Dirichlet B.C.: $G(x, y) = \Phi(y - x) - \Phi\left(\frac{|x|}{R}(y - \tilde{x})\right)$.
 $x, y \in B_R$.

- Other Planar Surface Boundary: possible by images...

Energy Method

- Uniqueness $-\Delta u = 0$ in U ; $u = 0$ or $\frac{\partial u}{\partial \nu} u = 0$ on ∂U .

$$-\int_U u \Delta u dx = \int_U |Du|^2 dx \Rightarrow Du \equiv 0, u = \text{const. in } U.$$

- Dirichlet's Principle

$u \in C^2(\bar{U})$ solves $-\Delta u = f$ in U and $u = g$ on ∂U



$I(u) = \min_{w \in \mathcal{A}} I(w)$, where $\mathcal{A} = \{w \in C^2(\bar{U}) | w = g \text{ on } \partial U\}$

and $I(w) = \int_U (\frac{1}{2} |Dw|^2 - wf) dx$, $w \in \mathcal{A}$, 'energy' function.

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Heat Equation

Heat equation $u_t - \Delta u = 0$.

Nonhomogeneous heat equation $u_t - \Delta u = f$.

$u : \bar{U} \times [0, \infty) \rightarrow \mathbb{R}$ unknown; $f : U \times [0, \infty) \rightarrow \mathbb{R}$ given.

- u : chemical concentration, temperature, etc evolution in time.
- Also arise in other areas, e.g., Brownian motion, etc.

A guiding principle:

statements for Laplace equation

→ more complicated ones for heat equation

1. Fundamental Solution

- If $u(x, t)$ solves $u_t - \Delta u = 0$, then
so does $u(\lambda x, \lambda^2 t)$, $\lambda \in \mathbb{R}$; so do $u_t, u_x, u_{tx}, \dots$

May let $u = v\left(\frac{|x|^2}{t}\right)$ for some v ;

or, let $u = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right)$ (s.t. $u(x, t) = \lambda^\alpha u(\lambda^\beta x, \lambda t)$)

Plug in, choose $\beta = \frac{1}{2}$; then let $v(y) = w(|y|)$;

choose $\alpha = \frac{n}{2}$: $u(x, t) = \frac{b}{t^{n/2}} e^{-\frac{|x|^2}{4t}}$

- Fundamental solution

$$\Phi(x, t) = \begin{cases} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}}, & x \in \mathbb{R}^n, t > 0, \\ 0, & x \in \mathbb{R}^n, t < 0. \end{cases}$$

$$\int_{\mathbb{R}^n} \Phi(x, t) dx = 1, \quad \forall t > 0.$$

$$\Phi_t - \Delta \Phi = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty);$$

$$\Phi(x, 0) = \delta_0(x) \quad \text{on } \mathbb{R}^n \times \{t = 0\}.$$

- Initial value problem (Cauchy problem)

$u(x, t) = \int_{\mathbb{R}^n} \Phi(x - y, t)g(y)dy$ solves

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \mathbb{R}^n \times (0, \infty); \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

(The 2nd: $\lim_{(x,t) \rightarrow (x_0,0)} u = g(x_0), x, x_0 \in \mathbb{R}^n, t > 0$.)

Infinite propagation speed:

if g bounded, continuous; $g \geq 0, \not\equiv 0$, then
 $u(x, t) > 0, \forall x \in \mathbb{R}^n, t > 0$.

· Nonhomogeneous Problem (Duhamel's Principle)

$u(x, t) = \int_0^t u(x, t; s) ds$ solves

$$\begin{cases} u_t - \Delta u = f, & \text{in } \mathbb{R}^n \times (0, \infty); \\ u = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

where $u(x, t; s) = \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy$
solves

$$\begin{cases} u_t(\cdot; s) - \Delta u(\cdot; s) = 0, & \text{in } \mathbb{R}^n \times (s, \infty); \\ u(\cdot; s) = f(\cdot, s), & \text{on } \mathbb{R}^n \times \{t = s\}. \end{cases}$$

2. Mean-value formula $u \in C_1^2(U_T)$ solves heat eq.

$$u(x, t) = \frac{1}{4r^n} \int_{E(x, t; r)} u(y, s) \frac{|x-y|^2}{(t-s)^2} dy ds,$$

for each $E(x, t; r) \subset U_T$.

$U \subset \mathbb{R}^n$ open bounded; $T > 0$ fixed.

$U_T = U \times (0, T]$: parabolic cylinder;

$\Gamma_T = \bar{U}_T - U_T$: parabolic boundary.

$$E(x, t; r) = \{(y, s) \in \mathbb{R}^{n+1} | s \leq t, \Phi(x-y, t-s) \geq \frac{1}{r^n}\}:$$

'heat ball' (analogy of ball for Laplace equation).

boundary: level set of $\Phi(x-y, t-s)$; (x, t) : center.

3. Maximum Principle & Uniqueness

- Maximum principle in bounded domain U :

$$u \in C_1^2(U_T) \cap C(\bar{U}_T), \quad u_t - \Delta u = 0 \text{ in } U_T.$$

$$\cdot \max_{\bar{U}_T} u = \max_{\Gamma_T} u;$$

$$\cdot U \text{ connected; } u(x_0, t_0) = \max_{\bar{U}_T} u, (x_0, t_0) \in U_T$$

$$\Rightarrow u \text{ constant in } \bar{U}_{t_0}. \quad (\text{Strong Max. Principle})$$

Infinite propagation speed: $u_t - \Delta u = 0$ in U_T ;
 $u = 0$ on $\partial U \times [0, T]$; $u = g \geq 0$ on $U \times \{t = 0\}$.
 $g > 0$ somewhere on $U \Rightarrow u > 0$ everywhere in U_T .

Uniqueness for Initial/boundary value problem:

$$\begin{cases} u_t - \Delta u = f, & \text{in } U_T, \\ u = g, & \text{on } \Gamma_T, \end{cases}$$

at most one solution $u \in C_1^2(U_T) \cap C(\bar{U}_T)$.

- Maximum principle in unbounded domain $\mathbb{R}^n$:

$u \in C_1^2(\mathbb{R}^n \times (0, T]) \cap C(\mathbb{R}^n \times [0, T])$ solves

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \mathbb{R}^n \times (0, T); \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

$|u(x, t)| \leq A e^{a|x|^2}$ on $\mathbb{R}^n \times [0, T]$, $A, a > 0$ constant.

Then, $\sup_{\mathbb{R}^n \times [0, T]} u = \sup_{\mathbb{R}^n} g$. (Max. Principle)

Idea: $v(x, t) = u(x, t) - \frac{\mu}{(T + \epsilon - t)^{n/2}} e^{\frac{|x-y|^2}{4(T+\epsilon-t)}}$;
in a bounded domain;
 $u(y, t) = v(y, t) + \mu * \text{bdd}$, $\mu \rightarrow 0$.

Uniqueness for Cauchy Problem: f, g continuous,

$$\begin{cases} u_t - \Delta u = f, & \text{in } \mathbb{R}^n \times (0, T); \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

at most one $u \in C_1^2(\mathbb{R}^n \times (0, T]) \cap C(\mathbb{R}^n \times [0, T])$
satisfying $|u(x, t)| \leq Ae^{a|x|^2}$, $A, a > 0$ constant.

Remark Consider $f = 0, g = 0$.

$u = 0$: the physical correct solution;

other solutions: infinite many; nonphysical;

grows very rapidly as $|x| \rightarrow \infty$.

4. Regularity and Local Estimates

$u \in C_1^2(U_T)$ solves the heat equation.

- $u \in C^\infty(U_T)$, even if u nonsmooth on Γ_T .
- For integer k, l , $\exists$ constant C_{kl} s.t.

$$\max_{C(x,t;r/2)} |D_x^k D_t^l u| \leq \frac{C_{kl}}{r^{k+2l+n+2}} \|u\|_{L^1(C(x,t;r))}.$$

for all cylinders $C(x, t; r/2) \subset C(x, t; r) \subset U_T$, and
for all heat equation solutions u in U_T .

- $x \mapsto u(x, t)$ analytic, for each $0 < t \leq T$.
 $t \mapsto u(x, t)$ NOT analytic in general.

Energy Method

- Uniqueness $u_t - \Delta u = f$ in U_T ; $u = g$ on Γ_T .
Equivalently, $f = 0$ and $g = 0 \Rightarrow u \equiv 0$.

Consider $e(t) = \int_U u^2(x, t) dx$, $0 \leq t \leq T$.

$$\frac{d}{dt}e(t) = \dots = - \int_U |Du|^2 dx \leq 0; \quad e(0) = 0, \quad e(t) \geq 0$$

$\Rightarrow e(t) = 0$, and $u = 0$ in U_T .

- Backwards Uniqueness

$$u \text{ and } \tilde{u} \text{ satisfy } \begin{cases} u_t - \Delta u = f, & \text{in } U_T, \\ u = g, & \text{on } \partial U \times [0, T], \end{cases}$$

$$u(x, T) = \tilde{u}(x, T), \quad x \in U \Rightarrow u \equiv \tilde{u} \text{ in } U_T.$$

$$\text{Let } w = u - \tilde{u}, \quad e(t) = \int_U w^2 dx.$$

$$\text{Then, } e'(t) \leq 0, \quad e(T) = 0.$$

$$\text{By Cauchy-Schwartz, } (e'(t))^2 \leq e(t)e''(t).$$

$$\Rightarrow \text{when } e(t) \neq 0, \quad f(t) = \log e(t) \text{ convex } (f'' \geq 0).$$

$$\Rightarrow e(t) \equiv 0, \text{ and } u \equiv \tilde{u}.$$

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Wave Equation

Wave equation $u_{tt} - \Delta u = 0$.

Nonhomogeneous wave equation $u_{tt} - \Delta u = f$.

$u : \bar{U} \times [0, \infty) \rightarrow \mathbb{R}$ unknown; $f : U \times [0, \infty) \rightarrow \mathbb{R}$ given.

- u : a simplified model for a vibrating string ($n=1$), a membrane ($n=2$), or an elastic solid ($n=3$); the displacement of some direction of x at time t . (Two initial conditions: displacement u and velocity u_t .)
- Quite different from heat equation: u generally not C^∞ , finite speed of propagation, etc.

Solution of (Homogeneous) Wave Equation

$n = 1$: direct; $n \geq 2$: spherical mean.

- $n = 1$, d'Alembert's formula

$$\begin{cases} u_{tt} - u_{xx} = 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u = g, u_t = h, & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases}$$

$$u(x, t) = \frac{1}{2}[g(x+t) + g(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

Factorize the partial differential operator;
two transport eqns, or change of variables.

- $u(x, t) = F(x+t) + G(x-t)$, for some F and G :
sum of two travelling waves.
- $g \in C^k$ and $h \in C^{k-1} \implies u \in C^k$
(not in general smoother!)

Heat equation:

instantaneous smoothing of the initial data;
infinite speed of propagation.

Wave equation: NOT.

Initial/boundary-value problem: reflection method.

$$\begin{cases} u_{tt} - u_{xx} = 0, & \text{in } \mathbb{R}_+ \times (0, \infty), \\ u = g, u_t = h, & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ u = 0, & \text{on } \{x = 0\} \times (0, \infty). \end{cases}$$

g and h given; $g(0) = h(0) = 0$.

u, g, h : odd extension to $\mathbb{R}$.

$$u(x, t) = \begin{cases} \frac{g(x+t) + g(x-t) + \int_{x-t}^{x+t} h(y) dy}{2}, & x \geq t \geq 0, \\ \frac{g(x+t) - g(t-x) + \int_{t-x}^{x+t} h(y) dy}{2}, & t \geq x \geq 0. \end{cases}$$

$h = 0$, two waves: left and right; reflected at 0.

- $n \geq 2$: by the spherical mean; by method of descent.

$$\begin{cases} u_{tt} - \Delta u = 0, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

For $x \in \mathbb{R}^n$, $U(x; r, t) = \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y, t) dS(y)$.

$G(x; r), H(x; r)$: spherical mean of g, h .

$$\begin{cases} U_{tt} - U_{rr} - \frac{n-1}{r} U_r = 0, & \text{in } \mathbb{R}_+ \times (0, \infty), \\ U = G, U_t = H, & \text{on } \mathbb{R}_+ \times \{t = 0\}. \end{cases}$$

Euler-Poisson-Darboux eqn. ($\xrightarrow{n \text{ odd}}$ 1-dim wave eqn.)

• $n = 3$ ($n \geq 2$ odd, similar by spherical mean):

$$\tilde{U} := rU \quad \begin{cases} \tilde{U}_{tt} - \tilde{U}_{rr} = 0, & \text{in } \mathbb{R}_+ \times (0, \infty), \\ \tilde{U} = rG, \tilde{U}_t = rH, & \text{on } \mathbb{R}_+ \times \{t=0\}, \\ \tilde{U} = 0, & \text{on } \{r=0\} \times (0, \infty). \end{cases}$$

$$(\tilde{U} := (\frac{1}{r} \frac{\partial}{\partial r})^{k-1} (r^{2k-1} U), \quad n = 2k + 1 \quad (k \geq 1).)$$

$$u(x, t) = \lim_{r \rightarrow 0+} U(x; r, t) = \lim_{r \rightarrow 0+} \frac{\tilde{U}}{r} = \dots$$

$$= \frac{1}{|\partial B(x, t)|} \int_{\partial B(x, t)} [th(y) + g(y) + Dg(y) \cdot (y - x)] dS(y)$$

(Kirchhoff's formula)

• $n = 2$ ($n \geq 2$ even, similar by method of descent):

$$\bar{u}(x_1, x_2, x_3, t) = u(x_1, x_2, t), \quad \bar{g}, \bar{h} \text{ similar,}$$

$$\begin{cases} \bar{u}_{tt} - \Delta \bar{u} = 0, & \text{in } \mathbb{R}^3 \times (0, \infty), \\ \bar{u} = \bar{g}, \bar{u}_t = \bar{h}, & \text{on } \mathbb{R}^3 \times \{t = 0\}. \end{cases}$$

$$u(x, t) = \frac{1}{2|B(x, t)|} \int_{B(x, t)} \frac{t^2 h(y) + t g(y) + t Dg(y) \cdot (y - x)}{(t^2 - |y - x|^2)^{1/2}} dy.$$

(Poisson's formula)

Huygens' principle:

$$n \geq 2 \text{ odd, } h, g, Dg \text{ on } \partial B(x, t) \Rightarrow u(x, t);$$

$$n \geq 2 \text{ even, } h, g, Dg \text{ on } B(x, t) \Rightarrow u(x, t).$$

Solution of Nonhomogeneous Wave Equation

$$\begin{cases} u_{tt} - \Delta u = f, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = 0, u_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Duhamel's principle:

$u(x, t) = \int_0^t u(x, t; s) ds$ where $u(x, t; s)$ solves

$$\begin{cases} u_{tt}(\cdot; s) - \Delta u(\cdot; s) = 0, & \text{in } \mathbb{R}^n \times (s, \infty), \\ u(\cdot; s) = 0, u_t(\cdot; s) = f(\cdot, s), & \text{on } \mathbb{R}^n \times \{t = s\}. \end{cases}$$

Energy Method

- Uniqueness U open bounded w. smooth boundary ∂U

$$\begin{cases} u_{tt} - \Delta u = f, & \text{in } U_T, \\ u = g, & \text{on } \Gamma_T, \\ u_t = h, & \text{on } U \times \{t = 0\}. \end{cases}$$

$$f, g, h = 0 \Rightarrow u = 0.$$

$$\text{Energy function } e(t) = \frac{1}{2} \int_U [u_t^2(x, t) + |Du(x, t)|^2] dx;$$

$$\frac{d}{dt} e(t) = \dots = 0 \Rightarrow u_t = 0, Du = 0;$$

$$u = 0 \text{ in } U \times \{t=0\} \Rightarrow u = 0 \text{ in } U_T.$$

- Domain of Dependence (Finite Propagation Speed)

$$u_{tt} - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0, \infty).$$

Fix $x_0 \in \mathbb{R}^n$, $t_0 > 0$, consider the cone

$$C = \{(x, t) | 0 \leq t \leq t_0, |x - x_0| \leq t_0 - t\}.$$

$$u = u_t = 0 \text{ on } B(x_0, t_0) \times \{t = 0\} \Rightarrow u = 0 \text{ in } C.$$

Energy
$$e(t) = \frac{1}{2} \int_{B(x_0, t_0 - t)} [u_t^2(x, t) + |Du(x, t)|^2] dy.$$

$$\frac{d}{dt} e(t) = \dots \leq 0; \quad 0 \leq e(t) \leq e(0) = 0;$$

$$u_t = 0, Du = 0; \quad u = u(x_0, 0) = 0 \text{ in } C.$$

例子或类型：解的表示或性质

输运方程

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热/扩散方程

波动方程

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Nonlinear First-Order PDEs

$$\begin{cases} F(Du, u, x) = 0, x \in U; \\ u = g, x \in \Gamma \subset \partial U. \end{cases}$$

Example $u_t + |Du|^2 = 0.$

$u(x; a)$ is called a *complete integral* in $U \times A$, if

- $u(x; a)$ solves $F(Du, u, x) = 0, x \in U \subset \mathbb{R}^n$ for each $a \in A \subset \mathbb{R}^n$;
- $\text{rank}(D_a u, D_{ax}^2 u) = n, x \in U, a \in A$, that is, “ u depends on all n independent parameters $a_1, \dots, a_n$ ”.

$\Downarrow$ news solutions from envelopes

$v(x) = u(x; \phi(x))$ is called the *envelope* of $\{u(\cdot; a)\}_{a \in A}$:

$u(x; a), x \in U \subset \mathbb{R}^n, a \in A \subset \mathbb{R}^m; a = \phi(x)$ satisfies $D_a u(x; a) = 0.$

$(D_x v(x) = D_x u(x; a) \text{ for } a = \phi(x).)$

Method of Characteristics

$$\begin{cases} F(Du, u, x) = 0, x \in U; \\ u = g, x \in \Gamma \subset \partial U. \end{cases}$$

The characteristic ODE:

$$z(s) = u(\mathbf{x}(s)), \quad \mathbf{p}(s) = Du(\mathbf{x}(s)),$$

$$\begin{cases} \dot{\mathbf{p}}(s) = -D_x F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \\ \quad - D_z F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \mathbf{p}(s), \\ \dot{z}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \cdot \mathbf{p}(s), \\ \dot{\mathbf{x}}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)). \end{cases}$$

Example (Linear) $\begin{cases} x_1 u_{x_2} - x_2 u_{x_1} = u \text{ in } U, \\ u = g \text{ on } \Gamma, \end{cases}$ where

$$U = \{x_1 > 0, x_2 > 0\}, \Gamma = \{x_1 > 0, x_2 = 0\} \subset U.$$

Example
$$\begin{cases} u_t + (\frac{u^2}{2})_x = 0 \text{ in } \mathbb{R} \times (0, \infty), \\ u = g \text{ on } \mathbb{R} \times \{t = 0\}. \end{cases}$$

$$(1) \quad g(x) = \begin{cases} 1, & x \leq 0, \\ 1 - x, & 0 \leq x \leq 1, \\ 0, & x \geq 1. \end{cases}$$

$$(2) \quad g(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

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Other Ways to Represent Solutions

- Separation of Variables
(Sturm-Liouville Theory)
- Transform Method (Fourier, Laplace)
- Similarity Solutions
- Nonlinear to Linear
- Asymptotics
- Power Series

Example (Laplace's equation in a square)

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad (x, y) \in (0, a) \times (0, a);$$

$$\phi(0, y) = \phi(x, 0) = \phi(x, a) = 0, \quad \phi(a, y) = T(y), \\ x, y \in (0, a).$$

$$\phi(x, y) = \sum_{n=1}^{\infty} (A_n e^{n\pi x/a} + B_n e^{-n\pi x/a}) \sin\left(\frac{n\pi}{a} y\right),$$

$$\text{where } A_n = -B_n = \frac{1}{a \sinh(n\pi)} \int_0^a dy T(y) \sin\left(\frac{n\pi}{a} y\right).$$

- More complicated boundary conditions: superposition!
- Separable operator and separable boundary conditions: similar.
E.g., heat and wave equation in rectangular domain, etc.

Example (Separation in Polar Coordinates)
(2-dim steady flow past circular barrier)

$$\Delta \Psi = 0, \quad \nabla \Psi(\infty) = \mathbf{U}, \quad \frac{\partial \Psi}{\partial r} \Big|_{r=r_0} = 0.$$

(Ψ : velocity potential; $\nabla \Psi$: velocity.)

Laplace in polar coordinate: $\Delta = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2}.$

$$\Psi = \mathbf{U} \cdot \mathbf{r} (1 + r_0^2/r^2): \quad R(r)\Phi(\varphi), \quad \Phi \text{ periodic.}$$

$$a + b \ln r + \sum_{n=1}^{\infty} \left[a_n r^n \cos n(\varphi - \alpha_n) + \frac{b_n}{r^n} \cos n(\varphi - \beta_n) \right],$$

where $a_n = b_n = b = 0$ ($n \geq 2$), $a = \text{constant}$,

$$a_1 = U, \quad b_1 = r_0^2 U, \quad \alpha_1 = \beta_1 = \text{angle of } \mathbf{U}.$$

- Laplace equation separates in 11 coordinates.

Sturm-Liouville Theory

Vector space:

$$V = \{y(x) | \alpha_1 y(0) + \alpha_2 y'(0) = 0, \beta_1 y(1) + \beta_2 y'(1) = 0\}$$

Linear operator: $\mathcal{L} = -\frac{d}{dx}(p(x)\frac{d}{dx}) + q(x)$, $0 < x < 1$,

self-adjoint w.r.t. inner product $\langle u|v \rangle := \int_0^\pi u^*(x)v(x)dx$, that is,
 $\langle \mathcal{L}u|v \rangle = \langle u|\mathcal{L}v \rangle$.

Linear Operator: $\tilde{\mathcal{L}} = a(x)\frac{d^2}{dx^2} + b(x)\frac{d}{dx} + c(x)$, $0 < x < 1$,
 $a(x) \neq 0, \forall x$, let $w(x) = \frac{1}{a} e^{\int \frac{b}{a} dx}$, then $\mathcal{L} = w\tilde{\mathcal{L}}$ self-adjoint.

$\tilde{\mathcal{L}}$ self-adjoint w.r.t. $\langle u|v \rangle_w := \int_0^\pi wu^*v dx$.

$$\tilde{\mathcal{L}}y = \lambda y \iff \mathcal{L}y = \lambda wy.$$

• Regular Sturm-Liouville Problems

$$\mathcal{L} = -\frac{d}{dx}\left(p(x)\frac{d}{dx}\right) + q(x), \quad 0 < x < 1,$$

$$\mathcal{L}y = \lambda wy, \quad (\tilde{\mathcal{L}}y = w^{-1}\mathcal{L}y = \lambda y)$$

$$\alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad \beta_1 y(1) + \beta_2 y'(1) = 0.$$

$p > 0, p', q, w > 0$ continuous on $0 \leq x \leq 1$ (regular).

(singular: true in $(0, 1)$, at least one fails at 0 or 1.)

- (0) Existence of eigenvalues/eigenfunctions.
- (1) All eigenvalues and eigenfunctions are real-valued.
- (2) Eigenfunctions from different eigenvalues are orthogonal w.r.t. $\langle u|v \rangle_w$.

(3) Eigenvalues are all simple, and form an infinite increasing sequence $\lambda_1 < \lambda_2 < \cdots < \lambda_n < \cdots$, moreover, $\lambda_n \rightarrow \infty$ as $n \rightarrow \infty$.

(4) $\{\phi_n\}_{n=1}^{\infty}$: normalized eigenfunctions;
 f, f' : piecewise continuous on $[0, 1]$.

Then, $\sum_{n=1}^{\infty} \langle \phi_n | f \rangle_w \phi_n = \frac{1}{2}(f(x+) + f(x-))$, $x \in (0, 1)$; moreover, if f is cont.

on $[0, 1]$ and $f(0) = f(1) = 0$ when $\alpha_2 = \beta_2 = 0$, $\sum_{n=1}^{\infty} \langle \phi_n | f \rangle_w \phi_n = f(x)$,
 $x \in [0, 1]$.

(5) For any square integrable function f (w.r.t. weight w), $\sum_{n=1}^{\infty} \langle \phi_n | f \rangle_w \phi_n$ converges to f in the mean square sense.

- **Other self-joint conditions, e.g.,**

- Periodic B.C.,

$$\mathcal{L} = -\frac{d^2}{dx^2}; \quad y(\pi) = y(-\pi), \quad y'(\pi) = y'(-\pi).$$

- Singular Sturm-Liouville: e.g.,

$$\mathcal{L} = -\frac{d}{dx}\left((1-x^2)\frac{d}{dx}\right), \quad -1 < x < 1; \quad y, y' \text{ bounded as } x \rightarrow -1^+, 1^-.$$

(0)(1)(2) ✓.

(3)(4)(5) not necessarily true.

- **Solve ODE via eigenfunction expansion**

$$\mathcal{L}y = \mu wy + f, \quad 0 < x < 1.$$

$$\alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad \beta_1 y(1) + \beta_2 y'(1) = 0.$$

Fredholm Alternative Theorem

either, nonhomogeneous problem: unique solution $\forall f$,

or else, homogeneous problem: $\exists$ nontrivial solutions.

Example (Helmholtz Equation, in 3 variables)

$$-\Delta u = k^2 u, \quad 0 < x, y, z < L; \quad u = 0 \text{ on the boundary.}$$

Possible only if $k^2 = (n_x^2 + n_y^2 + n_z^2) \frac{\pi^2}{L^2}$, $n_x, n_y, n_z \in \mathbb{Z}$.

The solution is $u = A \sin\left(\frac{n_x \pi}{L} x\right) \sin\left(\frac{n_y \pi}{L} y\right) \sin\left(\frac{n_z \pi}{L} z\right)$.

Eigenvalues of $-\Delta$ in a cube. (Self-adjoint.)

Example $u_t + H(Du) = 0$ in $\mathbb{R}^n \times (0, \infty)$.

Additive (instead of multiplicative) separation of variables:

$$u(x, t) = w(x) + v(t); \quad \dots \quad u(x, t) = a \cdot x - H(a)t + b, \forall a \in \mathbb{R}^n, b \in \mathbb{R}.$$

Transform Method (Fourier/Laplace)

Fourier Transform $u(x) \in L^1(\mathbb{R}^n)$,

Fourier transform:

$$\hat{u}(y) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{-ix \cdot y} u(x) dx, \quad y \in \mathbb{R}^n.$$

Inverse Fourier transform:

$$\check{u}(y) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix \cdot y} u(x) dx, \quad y \in \mathbb{R}^n.$$

- $u = (\hat{u})^\vee; \quad \int_{\mathbb{R}^n} u \bar{v} dx = \int_{\mathbb{R}^n} \hat{u} \bar{\hat{v}} dy.$
- $\widehat{c_1 u + c_2 v} = c_1 \hat{u} + c_2 \hat{v};$
 $\widehat{D^\alpha u} = (iy)^\alpha \hat{u}; \quad \widehat{u * v} = (2\pi)^{n/2} \hat{u} \hat{v};$

Laplace Transform $u(t) \in L^1(\mathbb{R}_+)$,

$$u^\#(s) = \int_0^\infty e^{-st} u(t) dt, \quad s \in \mathbb{R}_+.$$

- $(c_1 u + c_2 v)^\# = c_1 u^\# + c_2 v^\#;$
 $(u^{(k)})^\#(s) = s^k u^\#(s), \quad (u * v)^\#(s) = u^\#(s) v^\#(s).$
- $u(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} u^\#(s) ds$

Example (Fourier Transform, Cauchy problem)

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \mathbb{R}^n \times (0, \infty); \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

$\hat{u}(y, t)$: Fourier transform to u in variable x .

$$\begin{cases} \hat{u}_t + |y|^2 \hat{u} = 0, & \text{in } \mathbb{R}^n \times (0, \infty); \\ \hat{u} = \hat{g}, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

$$\hat{u} = \hat{g} e^{-|y|^2 t},$$

$$u = \frac{1}{(2\pi)^{n/2}} g * (e^{-|y|^2 t})^\vee = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{4t}} g(y) dy.$$

Example (Laplace Transform Heat equation)

$$\begin{cases} u_t - \Delta u = 0, & \text{in } U \times (0, \infty); \\ u = g, & \text{on } U \times \{t = 0\}. \end{cases}$$

$u^\#(x, s)$: Laplace transform to u in variable t .

Then, we have the resolvent equation for $u^\#$

$$-\Delta u^\# + su^\# = g \text{ in } U.$$

Solutions of special forms reflecting symmetries of PDE

- Travelling wave: $x, t \in \mathbb{R}$,
 $u(x, t) = v(x - \sigma t)$, for some speed σ and profile v .

Plane wave: $x \in \mathbb{R}^n, t \in \mathbb{R}$,

$$u(x, t) = v(y \cdot x - \sigma t),$$

wavefront normal to $y \in \mathbb{R}^n$, velocity $\frac{\sigma}{|y|}$, profile v .

- Invariant under dilation scaling:

$$u(x, t) = \lambda^\alpha u(\lambda^\beta x, \lambda t).$$

Pick $\lambda = \frac{1}{t}$, $u = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right)$, for some α, β, v .

Example (Exponential Solution) $u = e^{i(y \cdot x + \omega t)}$,

- $u_t - \Delta u = 0$: $\omega = i|y|^2$;
- $u_{tt} - \Delta u = 0$: $\omega = \pm|y|$;
- $u_t + u_{xxx} = 0$ ($n = 1$, Airy's equation): $\omega = y^3$.

Example (Soliton) Korteweg-de Vries (KdV) equation

$$u_t + 6uu_x + u_{xxx} = 0 \text{ in } \mathbb{R} \times (0, \infty)$$

(A model for surface waves in water.)

$$u(x, t) = v(x - \sigma t): \quad -\sigma v' + 6vv' + v''' = 0,$$

$$v, v', v'' \rightarrow 0 \text{ as } s \rightarrow \pm\infty \text{ ("solitary wave"),}$$

Remarkable: complete integrable, $v(s) = \frac{\sigma}{2} \operatorname{sech}^2 \left(\frac{\sqrt{\sigma}}{2} (s - c) \right)$

Example (Scaling) Porous medium equation

$$u_t - \Delta(u^\gamma) = 0 \text{ in } \mathbb{R}^n \times (0, \infty).$$

$$u \geq 0, \gamma > 0 \text{ constant.}$$

$$\text{Scaling invariant: } u(x, t) = \lambda^\alpha u(\lambda^\beta x, \lambda t). \quad \Rightarrow \quad u = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right);$$

$$\alpha + 1 = \alpha\gamma + 2\beta, \quad \alpha v + \beta y \cdot Dv + \Delta(v^\gamma) = 0;$$

$$v(y) = w(|y|), \alpha = n\beta, \quad w(r) = \left(b - \frac{\gamma-1}{2\gamma}\beta r^2\right)_+^{\frac{1}{\gamma-1}}, \quad b \text{ constant.}$$

$$u(x, t) = \frac{1}{t^\alpha} \left(b - \frac{\gamma-1}{2\gamma}\beta \frac{|x|^2}{t^{2\beta}}\right)_+^{\frac{1}{\gamma-1}}, \quad \alpha = \frac{n}{n(\gamma-1)+2}, \beta = \frac{1}{n(\gamma-1)+2}.$$

- The set $\{u > 0\}$: moves with finite propagation speed.

Example $u_t - a\Delta u + b|Du|^2 = 0.$

(quasilinear parabolic;
stochastic optimal control theory.)

$w = \phi(u)$: $w_t = a\Delta w - [a\phi''(u) + b\phi'(u)]|Du|^2;$

$a\phi'' + b\phi' = 0$, $\phi(z) = e^{-\frac{bz}{a}}$, (**Hopf-Cole transform**)

$w_t - a\Delta w = 0 \dots$

Example $u_t - au_{xx} + uu_x = 0$.
(Viscous Burgers' equation)

$$w(x, t) = \int_{-\infty}^x u(y, t) dy,$$

$$w_t - aw_{xx} + \frac{1}{2}w_x^2 = 0, \quad (\text{previous case})$$

....

Example (steady, 2-dim, irrotational fluid flow)

$$\begin{cases} (\sigma^2(\mathbf{u}) - (u^1)^2)u_{x_1}^1 + (\sigma^2(\mathbf{u}) - (u^2)^2)u_{x_2}^2 \\ \quad - u^1 u^2 (u_{x_2}^1 + u_{x_1}^2) = 0, \\ u_{x_2}^1 - u_{x_1}^2 = 0. \end{cases}$$

($\sigma(\cdot)$: given local sound speed; $\mathbf{u}$: unknown velocity field.)

$x^1 = x^1(u_1, u_2), x^2 = x^2(u_1, u_2)$: **hodograph transform**
(independent $\leftrightarrow$ dependent variables).

$$\begin{cases} (\sigma^2(\mathbf{u}) - u_1^2)x_{u_2}^2 + (\sigma^2(\mathbf{u}) - u_2^2)x_{u_1}^1 + u_1 u_2 (x_{u_2}^1 + x_{u_1}^2) = 0, \\ x_{u_2}^1 - x_{u_1}^2 = 0. \end{cases}$$

$x = Dz(u), z : \mathbb{R}^2 \rightarrow \mathbb{R}$: **method of potential functions**

$$(\sigma^2(\mathbf{u}) - u_1^2)z_{u_2 u_2} + (\sigma^2(\mathbf{u}) - u_2^2)z_{u_1 u_1} + 2u_1 u_2 z_{u_1 u_2} = 0. \quad \dots$$

Example $\operatorname{div} \left(\frac{Du}{(1+|Du|^2)^{1/2}} \right) = 0$: in dimension 2,
 $(1 + u_{x_2}^2)u_{x_1x_1} - 2u_{x_1}u_{x_2}u_{x_1x_2} + (1 + u_{x_1}^2)u_{x_2x_2} = 0$.
 (Minimal Surface equation)

Consider new independent variables: $p_1 = u_{x_1}(x_1, x_2)$, $p_2 = u_{x_2}(x_1, x_2)$,
 define $v(\mathbf{p}) = \mathbf{x} \cdot \mathbf{p} - u(\mathbf{x})$ **Legendre transform**.

$$(1 + p_2^2)v_{p_2p_2} + 2p_1p_2v_{p_1p_2} + (1 + p_1^2)v_{p_1p_1} = 0 \quad (\text{linear}) \quad \dots$$

- May NOT transform the boundary conditions easily.

Example $-\epsilon \Delta u^\epsilon + \operatorname{div}(u^\epsilon \mathbf{b}) = \delta_0$ in $\mathbb{R}^2$.

(**Singular perturbation** of $\operatorname{div}(u\mathbf{b}) = \delta_0$ in $\mathbb{R}^2$: adding small multiple of higher order term.)

(Transport and small diffusion.)

Characteristic curve C from 0 (unperturbed): $\mathbf{x}(t)$.

Close to C : $x = \mathbf{x}(t) + y\nu(\mathbf{x}(t))$, ν normal, new coordinate system (t, y) .

$0 < \epsilon \ll 1$: by intuition of diffusion, consider $z = \epsilon^{-1/2}y$, t ; $v^\epsilon = \epsilon^{1/2}u^\epsilon$,
one would expect that $z, v^\epsilon = O(1)$.

$$\dots, \quad v_t^\epsilon - v_{zz}^\epsilon + (\beta z v^\epsilon)_z + \frac{\dot{\sigma}}{\sigma} v^\epsilon = O(\epsilon^{1/2}).$$

Let $\epsilon \rightarrow 0$, suppose $v^\epsilon \rightarrow v$, then $v_t - v_{zz} + (\beta z v)_z + \frac{\dot{\sigma}}{\sigma} v = 0$.

$$u^\epsilon = \epsilon^{-1/2}(v + o(1)).$$

Example $u_{tt} - \Delta u = 0$ in $\mathbb{R}^n \times (0, \infty)$.

To understand highly oscillatory solutions, e.g.,

$$\begin{cases} u_{tt}^\epsilon - \Delta u^\epsilon = 0, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u^\epsilon = a(x)e^{\frac{ip(x)}{\epsilon}}, u_t^\epsilon = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Consider $u^\epsilon(x, t) = e^{\frac{ip^\epsilon(x, t)}{\epsilon}} a^\epsilon(x, t)$ (**geometric optics** formulation),

$$a^\epsilon((p_t^\epsilon)^2 - |Dp^\epsilon|^2) = \epsilon^2(a_{tt}^\epsilon - \Delta a^\epsilon).$$

let $\epsilon \rightarrow 0$, suppose $p^\epsilon \rightarrow p$, $a^\epsilon \rightarrow a \neq 0$, i.e.,

$u^\epsilon = (a + o(1))e^{\frac{i(p+o(1))}{\epsilon}}$, where $p_t \pm |Dp| = 0$ (Hamilton-Jacobi).

Example $a^{ij}(\cdot) : Q$ -periodic, Q unit cube in $\mathbb{R}^n$.

$$- \sum_{i,j=1}^n \left(a^{ij}(x/\epsilon) u_{x_i}^\epsilon \right)_{x_j} = f, \text{ in } U; \quad u^\epsilon = 0, \text{ in } \partial U.$$

(**Homogenization**: high-frequency oscillations in coefficients)

$\epsilon \rightarrow 0$, u^ϵ ? Assume $u^\epsilon \rightarrow u$.

Trick: try **two-scale expansion** (x : macroscopic; x/ϵ : microscopic)

$$u^\epsilon(x) = u_0(x, x/\epsilon) + \epsilon u_1(x, x/\epsilon) + \epsilon^2 u_2(x, x/\epsilon) + \dots$$

Primarily interested in u_0 .

$$\frac{1}{\epsilon^2} L_1 u_0 + \frac{1}{\epsilon} (L_1 u_1 + L_2 u_0) + (L_1 u_2 + L_2 u_1 + L_3 u_0) + \{\epsilon, \epsilon^2, \dots \text{ terms}\} = f,$$

$$L_1 u_0 = 0, \quad L_1 u_1 + L_2 u_0 = 0, \quad L_1 u_2 + L_2 u_1 + L_3 u_0 = f, \dots,$$

...

Consider k^{th} -order quasilinear PDE

$\sum_{|\alpha|=k} a_{\alpha}(D^{k-1}u, \dots, u, x) D^{\alpha}u + a_0(D^{k-1}u, \dots, u, x) = 0,$
 $x \in U$, given normal directional derivatives up to order
 $k-1$ along an $(n-1)$ -dim hypersurface $\Gamma \subset U$.

WLOG, $\Gamma \subset \{x_n = 0\} \cap U$; $u, \frac{\partial u}{\partial x_n}, \dots, \frac{\partial^{k-1}u}{\partial x_n^{k-1}}$ given on Γ .

Formal power series: to calculate all partial derivatives. On Γ ,

$$u \text{ (given)} \Rightarrow \frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_{n-1}}, \quad \frac{\partial u}{\partial x_n} \text{ (given)} \Rightarrow \frac{\partial^2 u}{\partial x_1 \partial x_n}, \dots, \frac{\partial^2 u}{\partial x_{n-1} \partial x_n}, \dots,$$

$(k-1)^{\text{th}}$ partial derivatives (given or calculated) $\Rightarrow k^{\text{th}}$ ones except $\frac{\partial^k u}{\partial x_n^k}$,

$\frac{\partial^k u}{\partial x_n^k}$: obtained from PDE;

k^{th} partial derivatives $\Rightarrow (k+1)^{\text{th}}$ ones, $\frac{\partial^{k+1}u}{\partial x_n^{k+1}}$ by (differentiating) PDE;

Convergence issue (existence of power series solution):

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球面平均及降维

包络法

分离变量法（施图姆-刘维尔理论）

傅里叶/拉普拉斯变换

变量替换，级数解，渐近（多尺度分析）

.....

分析方法

磨光法

均值公式及极值原理

能量法

变分原理

.....

Concluding Remarks

Take-home messages

1. PDE的复杂性：解的形式，对初值边值的依赖性，弱解的必要性，等等。
2. Laplace/Poisson方程，热传导（扩散）方程，波动方程：解的形式、性质、以及分析方法。
（一般而言：简单、基本、常见的方程。）
3. 一些分析PDE解的方法或思想：视情况而试。
（其他分析问题的方法的积累。）
4. Sturm-Liouville理论：自伴算子（厄密特算符）。